AC9M6SP02 • Year 6 Mathematics

Coordinates in Four Quadrants

Locate, transform and reason about points across the Cartesian plane

We are learning to…

Students plot ordered pairs with positive and negative coordinates, identify quadrants and explain coordinate changes under horizontal, vertical and symmetric movement.

Represent | Reason | Calculate | Interpret | Verify

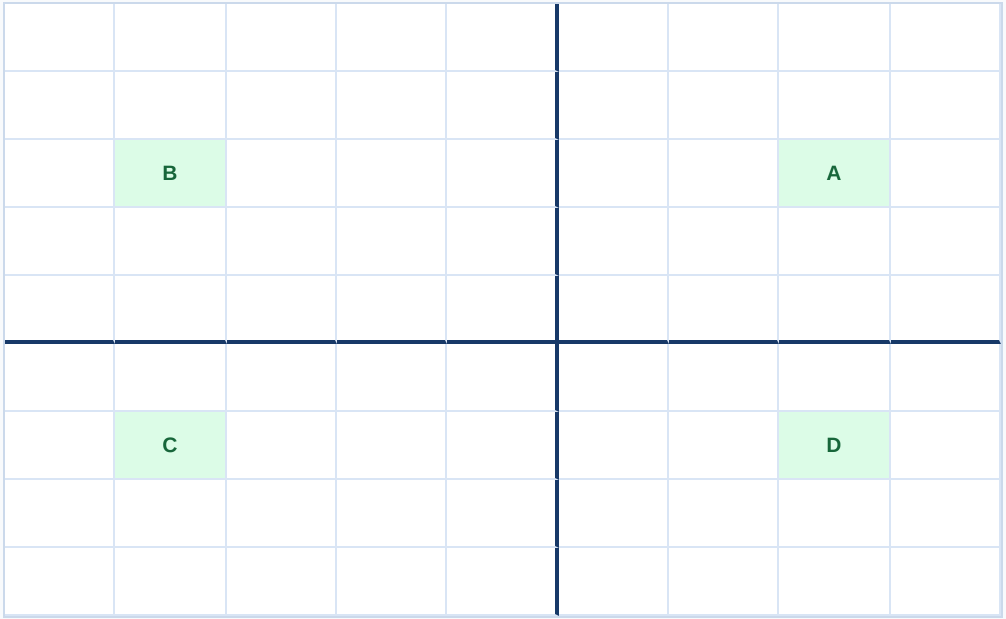
Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

locate points in the 4 quadrants of a Cartesian plane; describe changes to the coordinates when a point is moved to a different position

Plot one point in each quadrant



x first, then y

Quadrants are numbered anticlockwise from the upper right. The signs follow (+,+), (−,+), (−,−), (+,−).

Describe coordinate changes and symmetry

movement	coordinate rule
4 right	**$(x,y) \rightarrow (x+4,y)$**
3 down	**$(x,y) \rightarrow (x,y-3)$**
reflect in y-axis	**$(x,y) \rightarrow (-x,y)$**
reflect in x-axis	$(x,y) \rightarrow (x,-y)$
half-turn about origin	$(x,y) \rightarrow (-x,-y)$

State the transformation or displacement, then verify distance and orientation on the grid.

Build and annotate the model

Represent coordinates in four quadrants and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Coordinate order reversed:** Read x then y.
- **Negative direction ignored:** Left and down are negative on a standard plane.
- **Point on an axis assigned a quadrant:** Axis points are not inside a quadrant.
- **Reflection rule guessed:** Identify which coordinate changes sign.

Quick check

- Plot (−3,2).
- Name Quadrant IV.
- Translate a point.
- Reflect in y-axis.
- Identify an axis point.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.